

TAINAN AP, Taiwan

WMO: 467430

Lat: 22.950N

Lon: 120.206E

Elev: 19

StdP: 101.09

Time Zone: 8.00 (E08)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	12.0	5.0	5.4	12.8	7.1	6.3	13.6	11.1	15.2	9.9	15.4	4.7	10	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	33.2	27.6	32.9	27.4	32.2	26.9	28.6	32.1	28.1	31.6	27.7	31.2	5.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.4	31.0	27.1	22.9	30.7	26.8	22.5	30.4	92.3	32.5	90.7	31.7	88.8	31.3	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.1	DB	6.6	35.0	3.4	1.9	4.2	36.4	2.2	37.5	0.3	38.5	-2.1	39.9	(4)
(5)			WB	4.7	29.5	3.3	1.0	2.3	30.2	0.4	30.8	-1.4	31.3	-3.8	32.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	18.8	19.1	21.4	24.7	27.0	28.3	28.6	28.2	27.9	26.1	23.4	20.4	(6)
(7)		DBStd	4.16	2.56	3.13	2.89	2.51	1.65	1.37	1.20	1.50	1.24	1.68	2.10	2.40	(7)
(8)		HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	71	25	26	9	1	0	0	0	0	0	0	1	9	(9)
(10)		CDD10.0	5306	272	256	354	442	526	550	578	566	537	500	401	323	(10)
(11)		CDD18.3	2334	38	48	105	193	268	300	319	307	287	242	152	74	(11)
(12)		CDH23.3	22050	138	228	589	1517	2649	3468	3748	3446	3094	2040	846	287	(12)
(13)	CDH26.7	7780	8	31	111	438	913	1381	1554	1324	1153	657	178	32	(13)	
(14)	Wind	WSAvg	3.3	4.3	4.0	3.5	3.0	2.6	2.9	2.9	3.0	2.9	3.0	3.4	4.0	(14)
(15)	Precipitation	PrecAvg	1585	17	27	37	73	163	354	324	399	144	27	15	9	(15)
(16)		PrecMax	3112	103	108	173	300	416	1440	1038	1510	488	163	87	38	(16)
(17)		PrecMin	531	0	0	0	0	0	49	44	28	8	0	0	0	(17)
(18)		PrecStd	522	21	32	42	73	120	264	212	283	131	45	21	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.9	29.0	30.1	32.2	33.2	34.1	34.1	33.8	33.4	32.3	30.8	29.1	(19)
MCWB			21.6	22.6	24.1	26.1	27.2	27.8	28.1	28.3	27.5	26.1	24.6	22.9	(20)	
2%		DB	26.2	27.8	29.1	31.1	32.1	33.1	33.2	33.0	32.5	31.8	29.6	27.7	(21)	
		MCWB	20.5	21.9	23.7	25.6	26.4	27.5	27.6	27.8	27.0	26.0	24.2	21.9	(22)	
5%		DB	25.0	26.2	28.1	30.1	31.3	32.2	32.8	32.2	32.0	30.9	28.8	26.1	(23)	
		MCWB	19.8	20.9	22.7	24.9	26.1	27.0	27.4	27.3	26.7	25.7	23.4	21.0	(24)	
10%		DB	23.2	24.6	26.9	29.2	30.9	31.9	32.1	31.7	31.2	30.0	27.8	25.0	(25)	
		MCWB	18.7	20.0	21.8	24.4	25.9	26.9	27.0	27.1	26.4	24.9	22.5	20.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	24.0	25.5	27.0	28.2	29.0	29.0	29.2	28.7	27.2	25.7	24.1	(27)
MCDWB			26.7	27.5	29.0	31.0	31.6	32.6	33.0	32.5	31.9	31.3	29.5	28.1	(28)	
2%		WB	21.1	22.6	24.4	26.2	27.4	28.2	28.3	28.5	28.0	26.5	24.7	22.6	(29)	
		MCDWB	25.1	26.2	28.1	30.0	31.0	31.7	32.2	31.8	31.4	30.5	28.7	26.3	(30)	
5%		WB	20.2	21.5	23.6	25.6	26.7	27.7	27.9	28.0	27.4	26.0	23.8	21.6	(31)	
		MCDWB	24.1	25.2	27.0	29.3	30.3	31.2	31.6	31.3	30.9	29.7	27.6	25.1	(32)	
10%		WB	19.3	20.5	22.5	24.9	26.2	27.3	27.4	27.5	26.9	25.4	23.0	20.7	(33)	
		MCDWB	22.9	24.1	25.6	28.4	29.5	30.7	31.0	30.5	30.3	29.0	26.4	24.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	9.2	8.8	7.9	7.0	6.4	6.6	6.1	7.0	8.0	8.6	9.1	(35)
MCDBR			10.4	10.5	9.6	8.5	7.7	7.2	7.5	7.0	7.6	8.3	8.9	9.8	(36)	
MCWBR			5.9	6.1	5.3	4.7	4.0	3.6	3.8	3.7	3.9	4.2	4.5	5.5	(37)	
5% WB		MCDWB	9.9	10.3	8.9	8.1	7.2	6.7	6.9	6.5	6.9	7.7	8.3	9.3	(38)	
		MCWBR	6.1	6.4	5.5	5.0	4.3	3.8	3.9	3.9	4.1	4.1	4.6	5.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.507	0.551	0.665	0.656	0.542	0.497	0.502	0.555	0.595	0.600	0.525	0.524	(40)	
taud		2.005	1.918	1.687	1.754	2.079	2.235	2.213	2.036	1.900	1.839	2.022	1.972	(41)		
Ebn at Noon		746	742	678	696	775	805	801	761	721	690	724	709	(42)		
Edh at Noon		160	186	244	232	167	142	145	173	195	198	157	161	(43)		
(44)	All-Sky Solar Radiation	RadAvg	3.33	3.84	4.25	4.69	5.07	5.34	5.72	5.11	4.96	4.54	3.60	3.21	(44)	
(45)		RadStd	0.37	0.46	0.41	0.51	0.58	0.57	0.45	0.56	0.43	0.29	0.30	0.33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.24	N/A	N/A	N/A	+0.20	+0.24	N/A	N/A	+116	+110

Nomenclature: See separate page